20

Lecture 20

5G System Procedures

رویه‌های سامانهٔ نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 9, 12, 18 and 19

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-20-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Trace** initial registration including 5G-AKA, NAS security and subscription retrieval.
- **LO2** **Derive** the 5G-AKA verification values and key hierarchy at the level of their inputs and purpose.
- **LO3** **Trace** PDU session establishment across AMF, SMF, PCF, UPF and the RAN.
- **LO4** **Explain** UE- and network-triggered service request, RRC_INACTIVE and Xn/N2 handover.
- **LO5** **Estimate** procedure latency, AMF signalling load and handover forwarding buffers.

2 Prerequisite knowledge

This lecture uses the 5GC architecture and identities of Lecture 19, the SBI and NRF of Lecture 18, the EPS-AKA and EPC procedures of Lectures 9 and 12 (as a comparison) and the PFCP rules of Lecture 13. Procedures are presented in simplified form following TS 23.502 [6]; optional steps (roaming, interworking, many error cases) are omitted.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Registration	ثبت‌نام	Procedure that makes the UE known to the 5GC (initial, mobility, periodic).
5G-AKA	احراز هویت و توافق کلید ۵G	Authentication with home-network confirmation of RES*.
SEAF	تابع لنگر امنیتی	Security anchor in the serving network (co-located with the AMF).
NAS SMC	فرمان حالت امنیتی NAS	Activation of NAS ciphering and integrity.
PDU session establishment	برقراری نشست PDU	Creation of a session, its QoS flows and tunnels.
N1N2 message transfer	انتقال پیام N1N2	AMF service carrying SMF messages to UE (N1) and RAN (N2).
Service request	درخواست خدمت	Transition from CM-IDLE to CM-CONNECTED.
RRC_INACTIVE	حالت غیرفعال RRC	RAN-level suspended state with core context kept.
Path switch	تغییر مسیر	Moving the N3 tunnel to the target gNB after Xn handover.
ABBA	پارامتر ضدحمله به پایین‌برداری	Anti-bidding-down parameter bound into K_{AMF}.

3 Introduction and motivation

An architecture becomes concrete only through its procedures: the ordered exchanges that make a UE known to the network, secure its signalling, give it a data session and keep that session alive while the UE sleeps and moves. In the 5GC these procedures combine three kinds of signalling: NAS between UE and AMF (TS 24.501 [4]), NGAP between RAN and AMF (TS 38.413 [3]) and service operations between NFs (Lecture 18), plus PFCP towards the UPF.

Understanding the procedures is essential for troubleshooting (where did an attach fail?), for dimensioning (how many messages does each NF handle?) and for latency engineering (how long until the first packet flows?). This lecture traces the four central procedures and quantifies their cost.

4 Fundamental concepts

4.1 Types of registration

- **Initial registration**: the UE is RM-DEREGISTERED and registers for the first time (or after a long absence).
- **Mobility registration update**: the UE enters a tracking area outside its registration area, or its capabilities or requested slices change.
- **Periodic registration update**: on expiry of the periodic timer, to confirm reachability.
- **Emergency registration**: limited service for emergency calls.

4.2 Security anchors

5G-AKA [7] differs from EPS-AKA in one important respect: the *home network* confirms the authentication. The serving network's SEAF checks a hashed expected response (HXRES*) for a first verification, but the AUSF in the home network verifies RES* itself before releasing the anchor key K_{SEAF} and the SUPI. A visited network therefore cannot claim that a subscriber was present without proof – an answer to fraud concerns in roaming.

4.3 Session establishment roles

The AMF only relays session signalling: it selects an SMF and transports N1 (NAS SM) and N2 (NGAP resource set-up) messages for it. The SMF owns the session: it obtains policy from the PCF, selects and programs the UPF over N4, and asks the AMF to deliver the NAS accept to the UE and the resource request to the RAN in a single *N1N2 message transfer*.

5 Detailed technical explanation

5.1 Initial registration

The UE sends a Registration Request with its SUCI (or a 5G-GUTI if it has one), requested slices (NSSAI) and capabilities (Figure 1) [6]. The RAN selects an AMF (from the GUAMI in the 5G-S-TMSI or by load) and forwards the message in an NGAP Initial UE Message. If the UE gave a 5G-GUTI from another AMF, the new AMF fetches the context from the old AMF (Namf_Communication_UEContextTransfer). The AMF then authenticates the UE (5G-AKA, Section 5.2), activates NAS security, registers with the UDM (Nudm_UECM_Registration), retrieves the access and mobility subscription (Nudm_SDM_Get) and subscribes to changes, creates an AM policy association with the PCF, and, if needed, asks the NSSF to select slices. Registration Accept carries the new 5G-GUTI, the registration area and the allowed NSSAI; the UE acknowledges with Registration Complete.

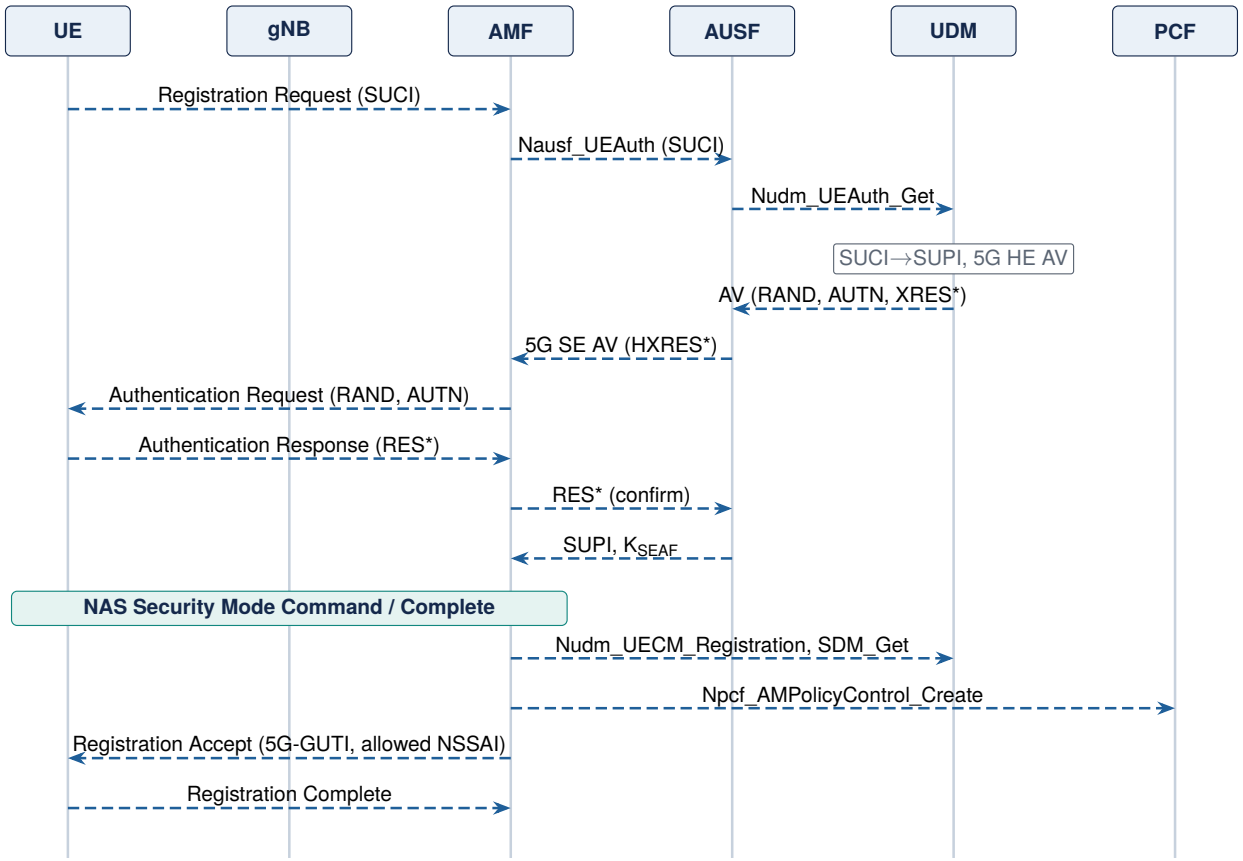


Figure 1. Initial registration with 5G-AKA (simplified) [6], [7].

5.2 5G-AKA

The UDM generates an authentication vector with the same functions f_1–f_5 as in Lecture 11, but derives 5G-specific values bound to the *serving network name* (SNN, of the form `5G:mnc<MNC>.mcc<MCC>.3gppnetwork.org`) [7]:

$$\mathrm{XRES}^* = \mathrm{KDF}\big(\mathrm{CK} \,\|\, \mathrm{IK},\ \mathrm{SNN},\ \mathrm{RAND},\ \mathrm{XRES}\big)\ \text{(128 bits)}, \tag{1a}$$

$$\mathrm{HXRES}^* = \mathrm{SHA\text{-}256}\big(\mathrm{RAND} \,\|\, \mathrm{XRES}^*\big)\ \text{(truncated to 128 bits)}, \tag{1b}$$

$$K_{\mathrm{AUSF}} = \mathrm{KDF}\big(\mathrm{CK} \,\|\, \mathrm{IK},\ \mathrm{SNN},\ \mathrm{SQN} \oplus \mathrm{AK}\big), \tag{1c}$$

followed by

$$K_{\text{SEAF}} = \text{KDF}(K_{\text{AUSF}}, \text{SNN}), \qquad K_{\text{AMF}} = \text{KDF}(K_{\text{SEAF}}, \text{SUPI}, \text{ABBA}),$$
$$K_{\text{gNB}} = \text{KDF}(K_{\text{AMF}}, \text{UL NAS COUNT}, \ldots).$$

The AUSF stores XRES* and gives the SEAF only HXRES*: the SEAF can check whether the hash of RAND and RES* computed as in Eq. (1b) matches, but cannot compute RES* itself, so it must forward RES* to the home network for the final decision.

5.3 PDU session establishment

The UE sends a PDU Session Establishment Request (with a new PDU session ID, DNN, S-NSSAI, session type and SSC mode) inside an UL NAS Transport message (Figure 2). The AMF selects an SMF (NRF discovery by DNN and slice) and calls Nsmf_PDUSession_CreateSMContext. The SMF retrieves session-management subscription data from the UDM, creates an SM policy association with the PCF (which returns PCC rules), selects a UPF and establishes the N4 session (PDRs, FARs, QERs, URRs), obtaining the uplink tunnel endpoint. It then calls Namf_Communication_N1N2MessageTransfer with the NAS accept (QoS rules, IP address) for the UE and the N2 resource set-up information (QoS flows and profiles, UPF tunnel endpoint) for the gNB. The gNB configures data radio bearers with RRC Reconfiguration and returns its downlink tunnel endpoint, which the SMF installs in the UPF with an N4 modification. QoS flows are now active in both directions.

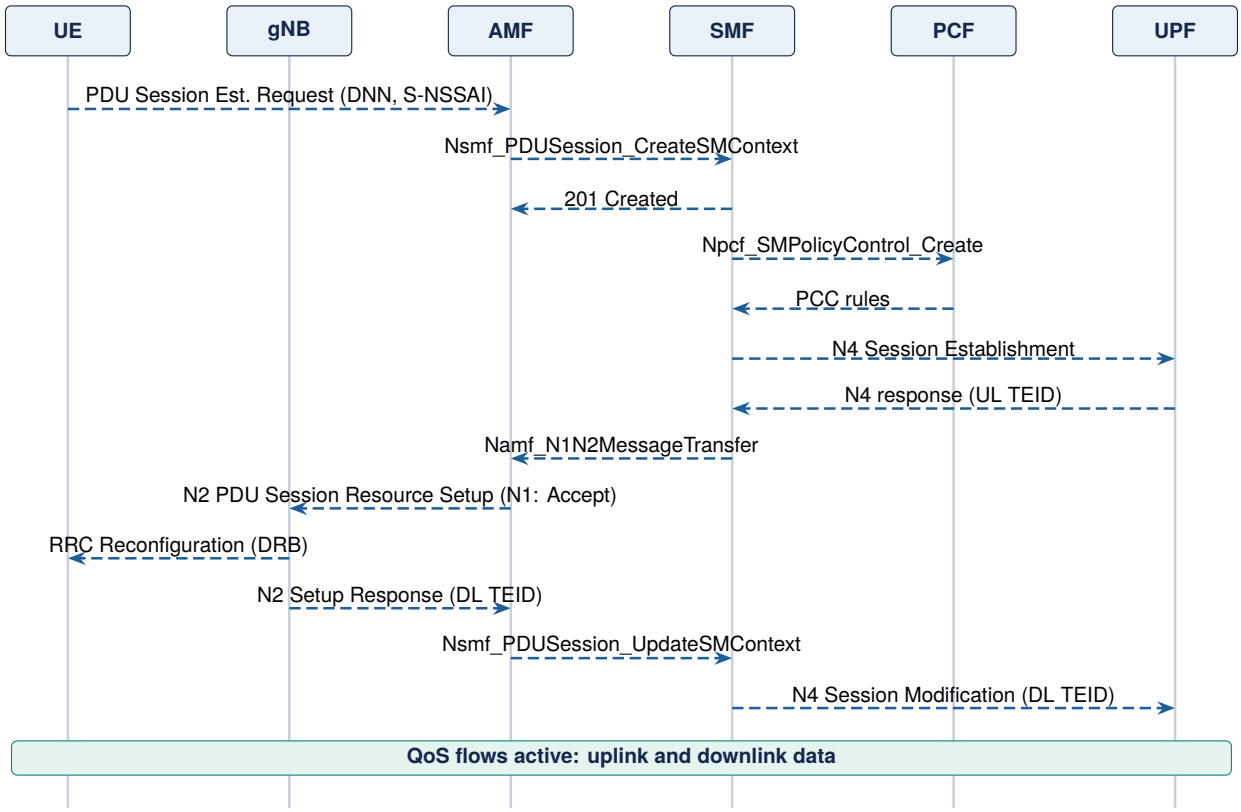


Figure 2. PDU session establishment (simplified) [6].

5.4 Service request and paging

A UE in CM-IDLE with uplink data sends a Service Request (with its 5G-S-TMSI) in RRC set-up; the AMF re-activates the user plane of the selected PDU sessions through the SMFs (UpdateSM-Context, N4 modification) and the RAN (N2 Initial Context Setup). For downlink data, the UPF buffers (or the SMF buffers, by FAR configuration) and notifies the SMF over N4; the SMF calls N1N2MessageTransfer; the AMF pages the UE in its registration area; the UE answers with a Service Request. A UE in RRC_INACTIVE skips most of this: the core context and N3 tunnels are kept, the RAN pages within its notification area and resumes the RRC connection locally, so no core signalling is needed unless the UE has moved out of the area.

5.5 Handover

With an Xn interface between gNBs, the source decides, prepares the target over Xn, commands the UE, forwards buffered data; the target then sends an N2 Path Switch Request so that the AMF and SMF move the N3 tunnel to the target, with end markers on the old path (Figure 3). Without Xn (or when the AMF or UPF must change), an *N2 handover* is prepared through the AMF(s), which also handles AMF relocation and inter-system handover to LTE.

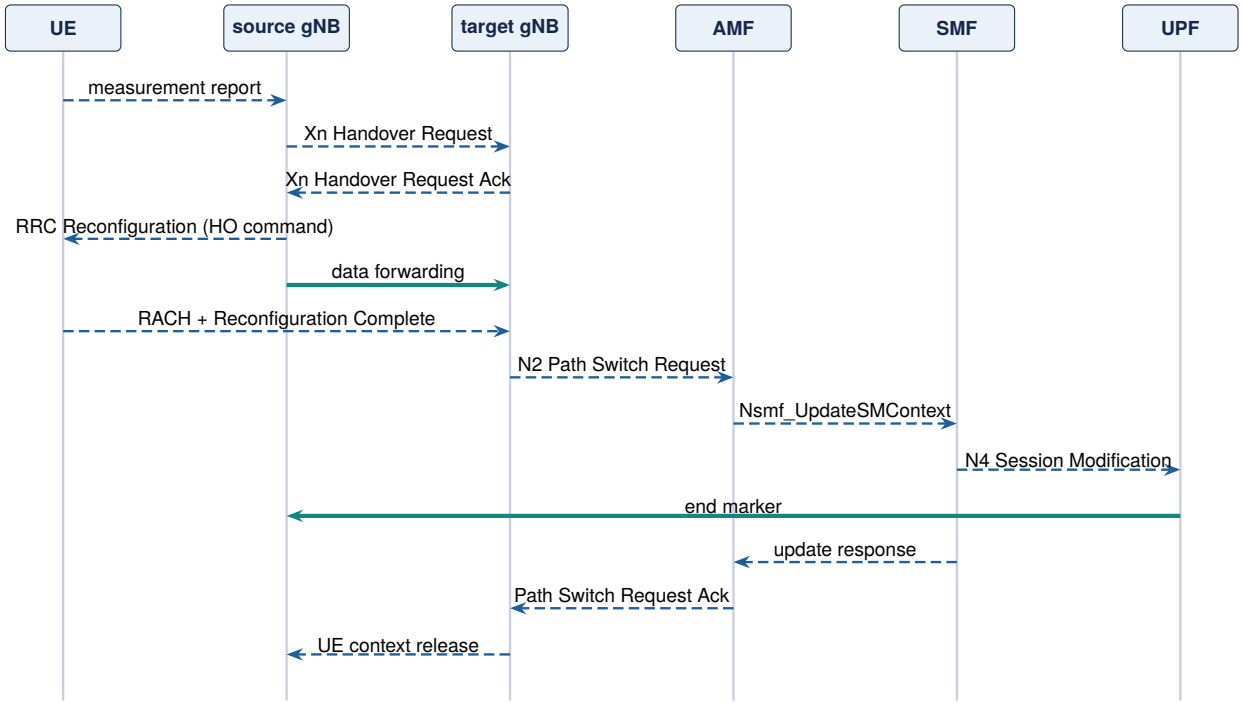


Figure 3. Xn handover with path switch (simplified).

6 Mathematical models

6.1 Procedure latency

Model. A procedure is a chain of sequential exchanges; parallel branches count only once. With n_{N} UE–AMF round trips of duration T_{N} (radio and RAN included) and n_{S} sequential core exchanges

of duration T_S,

$$T_{\text{proc}} \approx n_N\, T_N + n_S\, T_S. \tag{2}$$

Example. Initial registration (Figure 1) has about 6 UE–AMF round trips (registration, authentication, security mode, accept plus RRC) and 8 sequential SBI exchanges. With $T_N = 15\,\text{ms}$ and $T_S = 3\,\text{ms}$: $T_{\text{proc}} \approx 90 + 24 = 114\,\text{ms}$. The radio round trips dominate; core optimisation helps mainly when NFs are far apart or overloaded.

6.2 Signalling load at the AMF

Model. If procedure type k occurs r_k times per UE per hour and costs m_k messages at the AMF (NAS, NGAP and SBI counted), an AMF serving N UEs handles

$$\Lambda = \frac{N}{3600} \sum_k r_k\, m_k \quad \text{messages/s}. \tag{3}$$

Example. Illustrative values for $N = 10^6$: registrations 0.5/h at 14 messages, service requests 20/h at 6, Xn handovers 2/h at 4, PDU session establishments 0.2/h at 10: $\Lambda = 10^6 \times 137/3600 = 3.8 \times 10^4$ messages/s, of which service requests make up 88 % (Figure 4). If 70 % of idle-to-active transitions use RRC_INACTIVE (no core messages), Λ falls to 1.5×10^4 messages/s.

Interpretation. Frequent short sessions – smartphones and IoT devices waking for a few packets – dominate core signalling; RRC_INACTIVE is primarily a signalling-reduction feature.

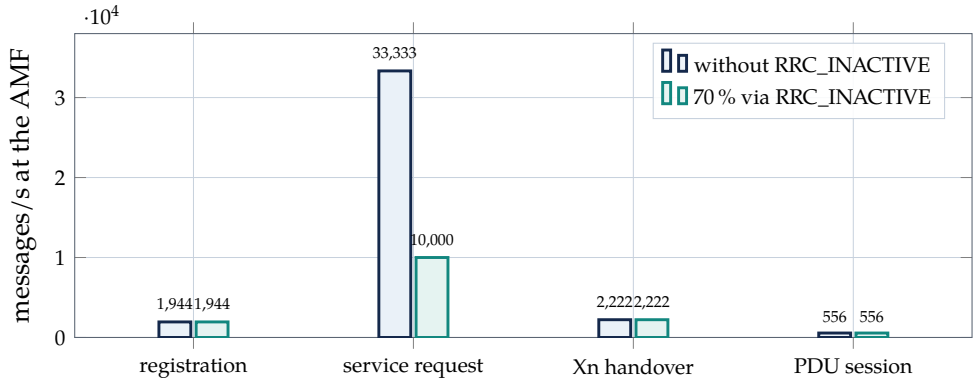


Figure 4. AMF message load per procedure type for the illustrative model of Section 6.2.

6.3 Forwarding buffer during handover

Model. Between the moment the source stops transmitting and the moment the path switch is complete, downlink data at rate R are forwarded over Xn and buffered at the target until the UE is ready. With interruption T_s,

$$B = R\, T_s. \tag{4}$$

Example. $R = 100\,\text{Mbit/s}$, $T_s = 20\,\text{ms}$: $B = 2\,\text{Mbit} = 250\,\text{kB}$ per UE. For an eMBB cell with several fast users the target needs megabytes of buffer; for URLLC, dual active protocol stack handover avoids the interruption altogether.

6.4 Authentication security

Model. An attacker without K who guesses RES* (128 bits) succeeds with probability 2^{-128} per attempt; one who has observed HXRES* must invert a 128-bit truncated hash. The expected number of attempts for a brute-force search over b bits is

$$\mathbb{E}[A] = 2^{b-1}. \tag{5}$$

Example. $b = 128$: 1.7×10^{38} attempts – infeasible; the security of 5G-AKA rests on the secrecy of K and the freshness of SQN, not on the length of the verification values.

7 Network architecture and diagrams

The figures of this lecture are *message sequence charts*: vertical lifelines for UE, RAN and NFs, horizontal arrows for messages. They abstract from transport (SCTP for NGAP, HTTP/2 for SBI, UDP for PFCP), from intermediate SCPs and from parallelism. In a real trace the same procedure appears as dozens of packets on several interfaces; reading traces efficiently means mapping them back to these charts.

8 Step-by-step procedures

Procedure 1 • Initial registration

1. UE → AMF: Registration Request (SUCI, requested NSSAI, capabilities) via RRC set-up and NGAP Initial UE Message.
2. AMF → AUSF → UDM: authentication request; the UDM de-conceals the SUCI and returns a 5G home-environment vector.
3. AMF (SEAF) → UE: Authentication Request; UE verifies AUTN and returns RES*; SEAF checks HXRES*; AUSF checks RES* and returns SUPI and K_{SEAF}.
4. NAS Security Mode Command/Complete; the AMF derives K_{AMF} and NAS keys.
5. AMF registers with the UDM and retrieves subscription data; creates the AM policy association; selects slices.
6. Registration Accept (5G-GUTI, registration area, allowed NSSAI); Registration Complete.

Procedure 2 • PDU session establishment

1. UE → AMF: PDU Session Establishment Request (DNN, S-NSSAI, type, SSC mode) in UL NAS Transport.
2. AMF selects an SMF and calls CreateSMContext; SMF fetches SM subscription, creates the SM policy association.
3. SMF selects a UPF and sends N4 Session Establishment with the rules; receives the UL tunnel endpoint.

4. SMF → AMF: N1N2MessageTransfer (N1: Accept with QoS rules and IP address; N2: QoS flows, UL TEID).
5. AMF → gNB: PDU Session Resource Setup; gNB configures DRBs and returns the DL TEID.
6. AMF → SMF: UpdateSMContext; SMF → UPF: N4 modification with the DL TEID.

9 Worked examples

Worked example 1 • Who checks what in 5G-AKA

Problem. A visited network's AMF receives RES* from the UE. Explain which checks it can perform and why it needs the AUSF.
Solution. The SEAF has HXRES* and RAND; it computes SHA-256(RAND ‖ RES*) (truncated) and compares with HXRES* (Eq. (1b)) – this filters wrong responses locally. It cannot derive XRES* (it lacks CK, IK) and does not receive the SUPI or K_{SEAF} until the AUSF has compared RES* with XRES*. The home network thus confirms the authentication.

Worked example 2 • Time to first packet

Problem. After power-on, a UE registers and then establishes a PDU session. The PDU session has 3 UE–AMF round trips (request, RRC reconfiguration with accept) and 6 sequential SBI/N4 exchanges. With the values of Section 6.1, when can the first packet flow?
Solution. Registration 114 ms; PDU session $3 \times 15 + 6 \times 3 = 63$ ms; total about 180 ms (plus cell search and system information acquisition, typically longer). Running the PDU session request piggybacked on registration (allowed for some cases) or pre-establishing sessions shortens it.

Worked example 3 • Dimensioning with RRC_INACTIVE

Problem. An AMF instance handles 2×10^4 messages/s. How many instances are needed for the load of Section 6.2 with and without RRC_INACTIVE, at 70 % target utilisation?
Solution. Without: $3.8 \times 10^4/(0.7 \times 2 \times 10^4) = 2.7 \rightarrow 3$ instances; with: $1.47 \times 10^4/1.4 \times 10^4 = 1.05 \rightarrow 2$ instances (plus redundancy in both cases).

10 Technology comparison

Always-on versus on-demand. The EPS attach always created a default bearer ("always-on IP"); 5G separates registration from session establishment. A device can register (e.g. for SMS or for a later session) without consuming user-plane resources, and sessions are created per DNN and slice as needed.

11 Key takeaways

Table 1. EPS and 5GS procedures compared.

	EPS	5GS
Attach / registration	Attach always creates a default bearer	Registration without session; PDU sessions separately
Authentication	EPS-AKA, decision in serving network	5G-AKA or EAP-AKA′, home confirmation
Identity	IMSI may be sent in clear	SUCI (concealed)
Session set-up	MME → S-GW → P-GW (GTPv2-C)	AMF relays; SMF drives PCF, UPF (SBI, PFCP)
Idle optimisation	ECM-IDLE only	CM-IDLE and RRC_INACTIVE
Handover path switch	MME → S-GW (S11)	AMF → SMF → UPF (SBI, N4)

Key takeaways

- Registration = identification (SUCI/5G-GUTI) + 5G-AKA + NAS SMC + UDM registration and subscription + policy + accept.
- 5G-AKA binds keys to the serving network name; SEAF checks HXRES*, the home AUSF confirms RES*.
- PDU session: AMF relays, SMF orchestrates PCF and UPF, N1N2 transfer delivers NAS accept and N2 set-up together.
- Service requests dominate AMF load; RRC_INACTIVE removes most of them from the core.
- Latency $\approx n_N T_N + n_S T_S$; load $= (N/3600)\sum r_k m_k$; forwarding buffer $= RT_s$.

12 Exercises

Exercise 1 • Level 1 – conceptual • Order of steps

Why must NAS security be activated before the AMF sends the Registration Accept? What could an attacker do otherwise?

Exercise 2 • Level 1 – conceptual • N1N2 message transfer

Explain why the SMF sends both the NAS accept and the N2 resource information to the AMF in one service operation rather than in two.

Exercise 3 • Level 2 – numerical • Latency

A mobility registration update needs 2 UE–AMF round trips and 4 sequential SBI exchanges (context transfer from the old AMF, UDM update, policy update, subscription retrieval). Compute its latency with $T_N = 15\,\text{ms}$ and $T_S = 3\,\text{ms}$, and with $T_S = 12\,\text{ms}$ if the old AMF is in a distant data centre.

Exercise 4 • Level 2 – numerical • Forwarding buffer

A cell has 5 users each receiving 300 Mbit/s when a train carries them all into a neighbouring cell simultaneously; the interruption is 30 ms. How much buffer does the target gNB need? What would reduce it?

Exercise 5 • Level 3 – analytical • IoT signalling

10^6 IoT devices per AMF send one small report every 15 minutes. Each report needs a service request (6 AMF messages) if the device returns to CM-IDLE, or none if it stays in RRC_INACTIVE (resumed in the RAN). Compute the AMF load in both cases, and discuss what RRC_INACTIVE costs the RAN instead.

Exercise 6 • Level 3 – analytical • Authentication in roaming

A visited network claims to have served a roaming subscriber for 2 hours and bills the home operator. With EPS-AKA and with 5G-AKA, what evidence does the home network have that the subscriber was really authenticated in the visited network? Relate your answer to Eq. (1).

Solutions

Solution to Exercise 1

The Registration Accept carries the new 5G-GUTI, the allowed slices and the registration area. Without integrity protection an attacker could modify these (e.g. restrict the UE's services or force it to reveal itself), and without ciphering the GUTI–SUCI link would be visible, undermining identity privacy. The security mode procedure also confirms the UE's security capabilities, preventing bidding-down attacks.

Solution to Exercise 2

The NAS accept and the RAN resource set-up belong together: the RAN must create radio bearers for the QoS flows the UE is told about, and the AMF can deliver both in one NGAP message (PDU Session Resource Setup with the NAS PDU inside). One operation keeps them consistent and saves a round trip; if the RAN rejects the resources, the AMF can report the failure to the SMF without the UE ever seeing an inconsistent accept.

Solution to Exercise 3

$2 \times 15 + 4 \times 3 = 42$ ms; with $T_S = 12$ ms: $30 + 48 = 78$ ms. Distant core NFs can dominate procedures with many sequential exchanges – an argument for locality-aware NF selection.

Solution to Exercise 4

$B = 5 \times 300 \times 10^6 \times 0.03/8 = 5.6$ MB. Reductions: shorter interruption (RACH-less or conditional handover, prepared target), dual active protocol stack (no interruption), or reducing the forwarded volume by letting TCP slow down; on trains, a moving relay or group handover helps.

Solution to Exercise 5

Reports: $10^6/900 = 1111\,\mathrm{s}^{-1}$. CM-IDLE: $1111 \times 6 = 6667$ AMF messages/s; RRC_INACTIVE: none for the reports (only periodic RAN notification area updates, also handled in the RAN). The RAN must keep the UE contexts and security state for all inactive UEs (memory) and handle RAN paging and resume; when a UE moves out of the RAN notification area, a RAN-based update is needed.

Solution to Exercise 6

With EPS-AKA the serving network compares RES with XRES itself; the home network only knows that it sent authentication vectors – it cannot prove the UE responded. With 5G-AKA the AUSF receives RES* and compares it with XRES* (Eq. (1a)); RES* can only be computed by a USIM holding K in response to that RAND and bound to that serving network name. The home network thus has cryptographic evidence that the subscriber was authenticated in that network at that time.

13 Further reading

The procedures are specified in TS 23.502 [6], with NAS in TS 24.501 [4], NGAP in TS 38.413 [3] and security (5G-AKA, key hierarchy, SUCI) in TS 33.501 [7]; the SBI operations used are in TS 29.500 [2] and the NRF in TS 29.510 [1]. RRC_INACTIVE and handover on the RAN side are in TS 38.300 [5]. Rommer et al. [8] walk through the procedures in detail.

References

[1] 3GPP, *5G system; network function repository services; stage 3*, TS 29.510, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29510.htm.

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[8] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.